university of groningen

Lorenzo Zanbelli

Practicum CFD 2026-2027

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Practicum 5
deadline: Tuesday, October 20th, 17:00

- Working in pairs (groups of max. 2 students) is allowed.
- Each practicum is closed with an oral discussion supported by a (short) written report of the findings. Make the report by pen/paper (readable) or with LaTeX, Word, etc.
- Always use Python or Matlab/Octave for the programs that you develop.
- Save your programs in separate files, such that it is easy to demonstrate them.
- The use of AI tools such as ChatGPT is not allowed!
- Be sure that you have discussed the exercises and programs before the deadline!

Navier-Stokes solver – general description

Required files: `cfdpract.f`, `cfd.in`, `liqdmenu.m`, `liqdplot.m`, `liqprint.m`, `xplot.m`.

Under Linux, the program `cfdpract.f` can be compiled with any Fortran compiler, e.g. by using one of the commands:

```
g77 -o cfdpract -O cfdpract.f
gfortran -o cfdpract -O cfdpract.f
```

which all generate the executable `cfdpract`. Then, the program can be run with the command `./cfdpract`.
Under Windows similar commands can be used in a command window. If a Fortran compiler is not available under UWP, the windows executable "cfdpract.exe" can be run directly, preferably in a black cmd window.
The program needs the input file `cfd.in`, which has to be adjusted to perform the different computations. As an example, the input file with parameter settings for Question 1 is included and described in the appendix.

The program writes output to screen and to output files `uvpf#.dat` and `config.dat`. The output to screen first gives the parameter values describing the problem. Then it gives each (user specified) time interval the time, div $\boldsymbol{u}$, the number of Poisson iterations and 3 horizontal velocities at the vertical center line of the domain (at the bottom, halfway, and at the top), following the statement in line 1823

```
WRITE(6,22) T,divmax,Iter,U(IMID,2),U(IMID,JM2Us/2),U(IMID,JM2Us)
```

The output file `config.dat` contains a description of the grid. The files `uvpf#.dat` contain the horizontal and vertical velocity and the pressure in each grid cell at a certain time. In the input file the user can specify whether the file `uvpf#.dat` is made each (user specified) time interval or only at the end of the computation.
The output files `uvpf#.dat` and `config.dat` are needed for the post-processing in Matlab. Typing the command `liqdmenu` in Matlab starts a menu with which the post-processing can be controlled. With the upper part of the menu contour plots of the velocities, pressure, vorticity or streamfunction can be made. With the lower part a movie of these plots at consecutive time steps can be made. With the middle part plots of the velocities or pressure along horizontal or vertical sections of the domain can be made.

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Part 1: treatment of div $\boldsymbol{u}^n$

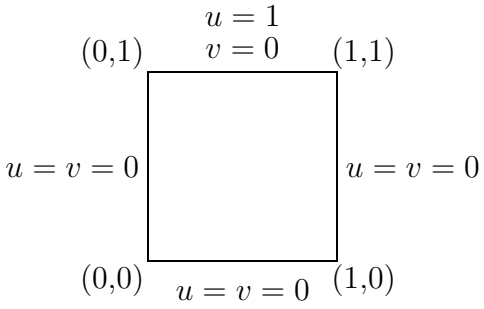
Description After explicit time-discretization and application of div $\boldsymbol{u}^{n+1} = 0$, the Navier-Stokes equations can be combined into a pressure Poisson equation and a velocity update equation

$$\text{div grad } p^{n+1} = \text{div } (\frac{\boldsymbol{u}^n}{\delta t} + \boldsymbol{R}^n), \tag{1a}$$

$$\boldsymbol{u}^{n+1} = \boldsymbol{u}^n + \delta t\ \boldsymbol{R}^n - \delta t \text{ grad } p^{n+1} \tag{1b}$$

in which $\boldsymbol{R} = -(\boldsymbol{u}.\text{grad})\boldsymbol{u} + \nu$ div grad $\boldsymbol{u}$ consists of convection and diffusion terms. The program `cfdpract.f` solves the Navier-Stokes equations in this way.
The term div $\boldsymbol{u}^n = 0$ in equation (1a) can be treated in two ways. The first way is to maintain this term. However, when the Poisson equation is solved exactly, div $\boldsymbol{u}^{n+1} = 0$. Therefore, the second way is to substitute this in (1a) for the next time step, i.e. substitute div $\boldsymbol{u}^n = 0$.

Flow configuration The *lid-driven cavity* problem is considered, i.e. the flow in a square cavity with constantly moving lid. This is a well known test case for Navier-Stokes solvers. The domain and boundary conditions of this problem are shown in the figure below.



Parameter settings in the input file We will solve the lid-driven cavity problem with $\nu = 0.01$, hence set `Nu=0.01` in the input file. Use a stretched grid with 16 cells in each direction: set `iMaxUs=jMaxUs=34`. Take smaller cells at the boundaries of the cavity: set `xpos=ypos=0.5` and stretching parameters `cx=cy=-0.95`. Discretize the convection terms with a central method: set `Alpha=0.0`.
Since the velocity at the top is prescribed: set `top=8`, `Uin=1.0` and `Vin=0.0` in the input file. Furthermore, since at the other boundaries we have no-slip conditions: set `left=2`, `right=2` and `bottom=2`.
Since the program was originally developed to compute unsteady flows, it does not contain a stopping criterion that checks whether a stationary solution is reached. Instead, the program stops when the simulation time is equal to a user prescribed time: take `TFin=25`. Set the time step at `DelT=0.025` and the maximum allowed CFL number at `CFLMax=0.5`. The program automatically adjusts the time step such that the CFL number is between 0.4*CFLMax and CFLMax.
In each time step the Poisson equation is solved with MILU. Use in the input file `PrtDt=1` and `uvp=0`, i.e. every second output is written to the screen and the velocity field is written to the file `uvpf0001.dat` at the end of the simulation.

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Question 1: accurate steady solution

First the 'exact' solution is computed, i.e. the discrete steady solution on the given grid. Hereto, set `Divuv=1` in the input file (i.e. retain div $\boldsymbol{u}^n$ in the Poisson equation), set `StrtP=1` (i.e. use the pressure of the previous timestep as initial guess for the iterations at the new time step) and use $\varepsilon = 10^{-3}$.
(a) Observe that the solution has converged in time, and write the results at $t = 25$ in the table below.

it has converged

time	div $\boldsymbol{u}$	# it.	u bottom	u mid	u top
25	$0.95367e^{-5}$	1	-0.0072	-0.22023	0.81385

(b) Estimate the maximum Peclet number in $x-$direction, and check the stability limits $\eta < d < 1$ (focus on $x-$direction).

Question 2: div $\boldsymbol{u}^n = 0$ substituted

Set `Divuv=0` in the input file. In each of the following questions, put the output of the program at $t = 20, 21, \ldots, 25$ in a table with the format given below.

time	div $\boldsymbol{u}$	# it.	u bottom	u mid	u top
20					
21					
22					
23					
24					
25					

(a) Use an accuracy $\varepsilon = 10^{-1}$ for the Poisson solver and use $p = 0$ as initial solution (set `StrtP=0`), i.e. start the Poisson solver from scratch

1) Look what happens with div $\boldsymbol{u}$ in time.
2) Make plots of the solution. Use the post-processing menu in Matlab to view the flow and use the upper part of the menu to make a surface plot of the streamfunction. Also have a look at the absolute velocity (`crange` [0, 0.1]) and the vertical velocity. Repeat the simulation with `TFin=100`. Does the velocity field converge?
3) Set (temporarily) `DelT=0.0025` and `CFLMax=0.05` in the input file and repeat the simulation (with `TFin=25`). Look what happens with the term div $\boldsymbol{u}$ in time. Compare this with the previous behaviour of div $\boldsymbol{u}$ and explain the results. Does $\delta t \to 0$ improve the solution?

(b) Use for the Poisson solver an accuracy $\varepsilon = 10^{-3}$ and again use $p = 0$ as initial solution (set `StrtP=0`). Look what happens with the term div $\boldsymbol{u}$ in time and make various plots of the solution as in the previous question. The solution has improved, but is it really accurate?

(c) Use for the Poisson solver again an accuracy $\varepsilon = 10^{-3}$, but now use the solution from the previous time step as initial solution (set `StrtP=1`). Look once more what happens with the term div $\boldsymbol{u}$ in time. Are you satisfied now?

(d) Use for the Poisson solver an accuracy $\varepsilon = 10^{-1}$ and again the solution from the previous time step as initial solution. Once again, describe the behaviour of div $\boldsymbol{u}$ in time.

Question 1

Lorenzo Campbell:

The first thing to consider is the maximum Peclet number is related to the maximum Δx and maximum u. Given $\Delta x = 1/32 \approx 0.03125$ is the mean mesh size. The main occur around the central location and is approximately 0.09. The maximum u is obtained at the top $u_{top} = 0.8135$ which we can bound to $u = 1$.

The Peclet number in the center is

$$P_{TOP} = \frac{0.8135 \cdot 0.09}{0.01} \approx 3.254$$

while the average ($\Delta x \approx 0.03125$)

$$P_{AV} \approx 2,59$$

Using $u \equiv 1$ then $P_{TOP} = 4$ & $P_{AVE} = 3.125$.
This implies that P is bounded above by 4 where the true actual maximum is around 3.254.

In all this cases $P > 2$ meaning the wiggle-free condition is not satisfied, but the problem is really close to.

The next step is check the stability $\eta < d < 1$

$$\eta = \frac{u \Delta t}{\Delta x}, \quad d = \frac{2\nu \Delta t}{\Delta x^2}$$

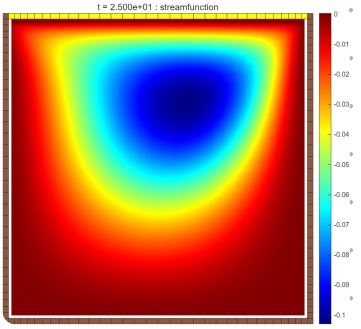
Using Avege time step of 0.0081 we get
$\Delta x \equiv 0.09$
$u \equiv u_{TOP} \Rightarrow \eta_{TOP} \approx 0.1648 \quad d \approx 0.05$

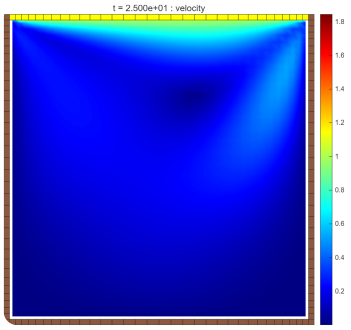
Clearly $\eta \geq d$ is not satisfied but $\eta^2 \approx 0.027 < d$ is. Which is the stricter condition we have in the lecture notes. To have also using $\Delta x = 0.03125$ the $\eta < d$ was not satisfied but yes $\eta^2 < d$.

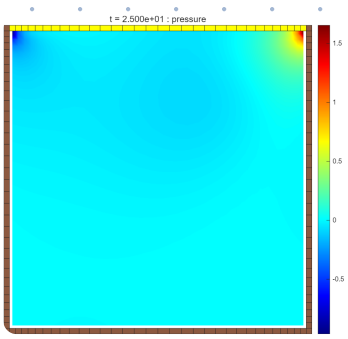
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Question 2

Before to start, let's plot pressure, absolute velocity and stream function for case describe in Q1.

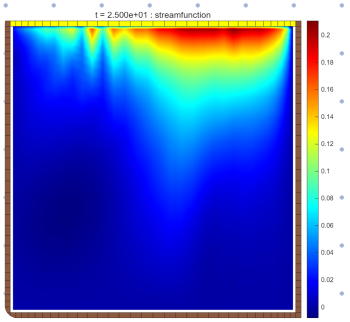


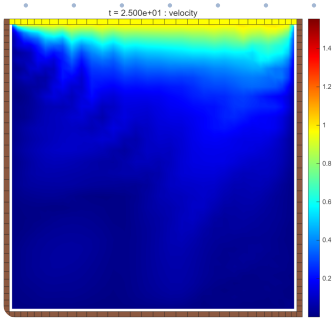




this we will use them as reference.

(a) here the plots and the table

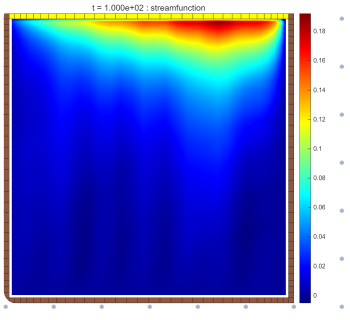


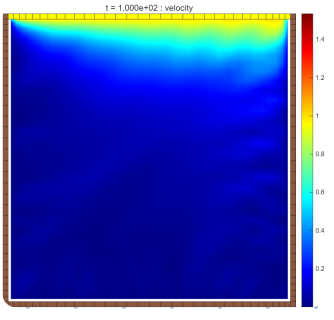


time	div u	# it.	u bottom	u mid	u top
20	$0.49596e^{+2}$	2	0.007	0.0909	0.84522
21	$0.49483e^{+2}$	2	0.00717	0.0937	0.84508
22	$0.49387e^{+2}$	2	0.00736	0.09603	0.84467
23	$0.49356e^{+2}$	2	0.00756	0.09736	0.84477
24	$0.49688e^{+2}$	1	0.00799	0.09241	0.84723
25	$0.49999e^{+2}$	2	0.00850	0.08329	0.84802

the pressure plot is omitted because completely off which is clear from the streamfunction plot.

By increasing the time to T=100 the result become.





Clearly the result is not converging to Q1 but it is possible to notice a reduction of vortices. the solution is getting closer to Q1.

In both cases the div u is big. From the solution we get t=1 ≈ 30.7 —> t=10 ≈ 48.5 —> t=25 ≈ 50 —> t=100 ≈ 49.7 Meaning the div does not converge.

With Stop=0 & $\varepsilon = 1e^{-1}$ the poisson equation is solved only very roughly and is restarted from p=0 at every time step. Meaning, the pressure correction is unable to enforce continuity. the div u increase and stabilize around $0.5e^2$ the stream function the velocity plots still show a dominant recirculation cell and t=100 is approaching a steady pattern, but not a physical one.

The fact that t=100 seem converge Lorenzo Tanbelli
is because it is reaching a steady state
$u_t = 0$

time	div u	# it.	u bottom	u mid	u top
20	$0.5008e^{+2}$	2	0.0061	.08696	.84501
21	$0.49946e^{+2}$	2	.00673	.08967	.84517
22	$0.4981e^{+2}$	2	-.00689	.09229	.84516
23	$0.4970e^{+2}$	2	.00697	.09430	.84505
24	$0.4962e^{+2}$	2	.00711	.09539	.84522
25	$0.4977e^{+2}$	1	.00746	.09010	0.84926

Dt=.0025
CFLMAX=.05

$\delta t \to 0$ does not improve the solution because the div error is not mainly caused by the time discretization but by the pressure correction. Recall that the projection method is

① Compute intermediate velocity
② Solve poisson equation
③ Correct the velocity s.t $\nabla \cdot u^{n+1} = 0$

} => If poisson is solve inaccurately $\epsilon = 1e^{-1}$ then it cannot improve the solution.

projection step

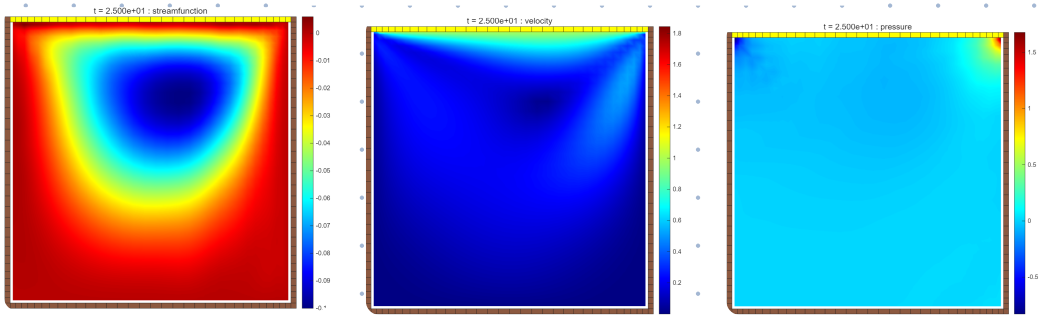
$$u^{n+1} = u^* - \Delta t \nabla p$$

$$\Rightarrow \quad \nabla \cdot u^{n+1} = \nabla u^* - \Delta t \nabla^2 p \qquad \text{but with } p = p_{exact} + e_p$$

$$\Rightarrow \quad \nabla \cdot u \propto T \nabla^2 e_p$$

(b) here figures and table

time	div u	# it.	u bottom	u mid	u top
20	$0.46135e^{+1}$	15	-0.00615	-0.21218	0.81338
21	$0.48922e^{+1}$	15	-0.00609	-0.21169	0.81302
22	$0.50701e^{+1}$	15	-0.00609	-0.21120	0.81285
23	$0.52934e^{+1}$	15	-0.00598	-0.21071	0.81268
24	$0.55137e^{+1}$	15	-0.00593	-0.21023	0.81250
25	$0.57329e^{+1}$	15	-0.00586	-0.20979	0.81231

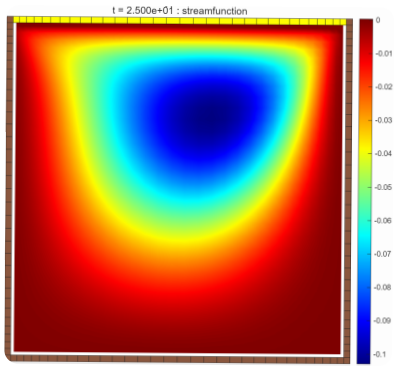


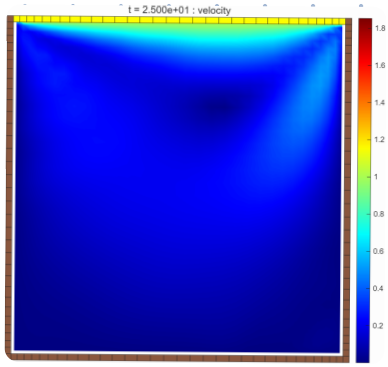
Starting from $p=0$ clearly implies more steps. In this case the decrease $\epsilon = 1e^{-3}$ improves the solution by 2 order of magnitude but still having div u increase over t makes sense over the theory ($\nabla \cdot u \propto T \nabla^2 e_p$).

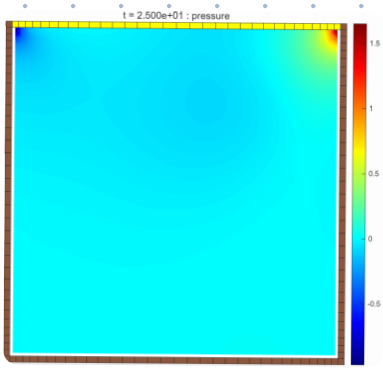
$\to 0 \Rightarrow \nabla \cdot u \to 0$

Accurate incompressible-flow behavior. Lorenzo Zambelli

(c) Figure & table







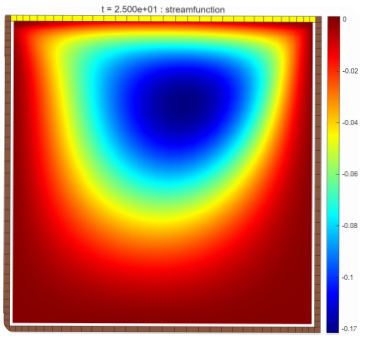
→ less step because $p^n \approx p^{n+1}$

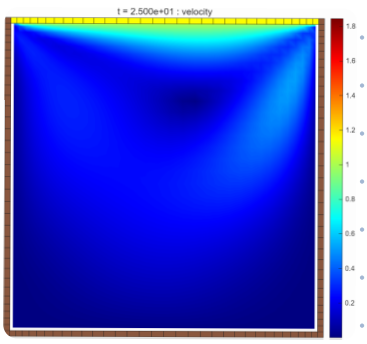
time	div u	# it.	u bottom	u mid	u top
20	0.19533	1	-0.00719	-0.21927	0.81349
21	0.19535	〃	〃	〃	〃
22	0.19537	〃	〃	〃	〃
23	0.19540	〃	〃	〃	〃
24	0.19542	〃	〃	〃	〃
25	0.19544	〃	〃	〃	〃

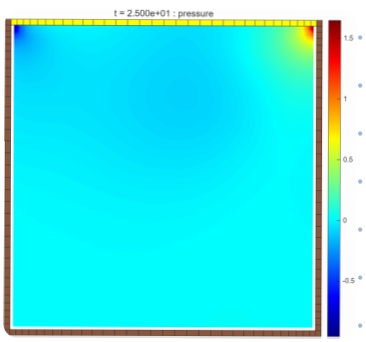
→ still around the initial div u.

Using step=1 increase the accuracy of div u by 1 order of magnitude. The solution still look very similar, but this time div u has a better value closer to div $u=0$. Although we can still see div u to increase over time, which is true sense. So this solution is STATIONARY for small t.

(d)







time	div u	# it.	u bottom	u mid	u top
20	.14126e-1	1	-.00860	-.26272	.81750
21	.14127e-1	1	〃	〃	〃
22	.14128e-1	1	〃	〃	〃
23	.14128e-1	1	〃	〃	〃
24	.14129e-1	1	〃	〃	〃
25	.14130e-1	1	〃	〃	〃

By increasing ε to $1e^{-1}$ the div u increase by 4 order of magnitude. The solution has good appearance but not really div u and of course worse with respect to $\varepsilon = 1e^{-3}$. But it is considerably better than starting from $p=0$ since $p^n \approx p^{n+1}$.

Question 3: div $\boldsymbol{u}^n$ retained

(a) Set `Divuv=1` in the input file. Use an accuracy $\varepsilon = 10^{-1}$ and $p = 0$ as initial solution for the Poisson solver.

1) Look what happens with div $\boldsymbol{u}$ in time. How do you like the solution?

2) Now we will consider what happens with the divergence when we let $\delta t \to 0$. Set (temporarily) `DelT=0.0025` and `CFLMax=0.05` in the input file. Repeat the simulation. Look what happens with the term div $\boldsymbol{u}$ in time. Compare with the previous results and explain the differences.

(b) Use an accuracy $\varepsilon = 10^{-3}$ and $p = 0$ as initial solution for the Poisson solver. Look what happens with the term div $\boldsymbol{u}$ in time. Compare the results with those from a), and explain the differences.

(c) Use again an accuracy $\varepsilon = 10^{-3}$ and use the solution from the previous time step as initial solution (set `StrtP=1`). Monitor div $\boldsymbol{u}$ in time, and make plots of the solution. Explain the results.

d) Use $\varepsilon = 10^{-1}$ and use the solution from the previous time step as initial solution. Monitor div $\boldsymbol{u}$ and plot the solution. How do you like the results?

Question 4 We have seen that the various approaches give quite different results. In practical applications one is often satisfied when the final solution has an accuracy of two digits. Keeping this in mind, compare the results from the previous questions and give your opinion about which approache(s) give(s) acceptable results.

Question 3

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(a) over time DIV u ≈ 0.53 while the velocities are converged to −0.00718, −0.21865 & 0.81338.

In this case $DIV\,u^{(n+1)} = \delta t\, \epsilon^{(n+1)}$, hence $\delta t \to 0$ DIV u → 0 which is what we see in the result where the DIV u ≈ $0.81970\epsilon^{-1}$. the velocities are the same as initial case.

In other words, in Q2 Δt has little effect because the dominant error come from the intermediate poisson solution. Here, DIVU=1 feeds the divergence back into the pressure equation, decreasing the time step now has a stronger effect.

(b) t=1 DIV u ≈ $1.83\epsilon^{-3}$, t=10 DIV u ≈ $1.89\epsilon^{-3}$, t=25 DIV u ≈ $1.89\epsilon^{-3}$

the DIVERGENCE remain practically constant. Decreasing the poisson tolerance from 10^{-1} to 10^{-3} decrease the divergence by almost 3 order of magnitude. But since it start every time from p=0 it is much lower then STARTP=1.

(c) Plots made before. Using these settings gives the best solution as can because using the pressure fields from the previous time step & the poisson equation has the DIV term in it. So the poisson eq see the continuity error, any non-zero DIV u acts as a source term and the pressure tries to eliminate it.

(d) As t → 25 DIV u ~ 10^{-4} & 10^{-5}. Since we have ρ=1 & $\rho^n \approx \rho^{n+1}$ even increasing ε to $1\epsilon^{-1}$ is sufficient. the solver in this case perform about 5 iterations per time step.

In other words, a good initial step is more important than convergence tolerance. the resulting velocities are identical to $\epsilon = 1\epsilon^{-3}$.
the solution are SATISFACTORY.

Question 4

Settings	DIV u
DIVU=0, $\epsilon = 10^{-1}$, STARTP=0	O(50)
" $\epsilon = 10^{-3}$, "	O(1-10)
" " , STARTP=1	$O(10^{-1})$

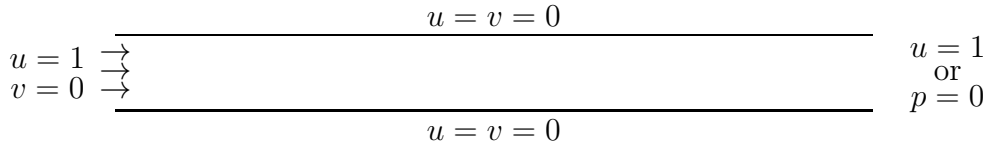
Settings	DIV u
DIVU=1, $\epsilon = 10^{-1}$, STARTP=0	$O(10^{-1})$
" , $\epsilon = 10^{-3}$, "	$O(10^{-3})$
" , " , STARTP=1	$O(10^{-5})$
" , $\epsilon = 10^{-1}$, "	$O(10^{-5})$

Clearly introducing DIV u into the poisson eq is essential to have good, acceptable DIV u. Using the previous pressure as initial step helps the solver, to reach better & faster results. Having a strict tolerance on the poisson is not necessary and loose ε can still give good results.
For practical application is to use DIVU=1 & STARTP=1 while keeping a loose ε.

Part 2: influence of boundary conditions

Description We will now consider the choice of boundary conditions at in- and outflow openings. The inflow boundary usually does not lead to too many problems. However, the outflow boundary is much more problematic. When at an outflow boundary the normal velocity is prescribed then a boundary layer may be created.

Flow configuration Consider the flow in a channel of length 10 and height 1. At the left (inflow) boundary the fluid flows into the channel with horizontal velocity $u = 1$. At the upper and lower boundary of the channel we impose no-slip conditions. Two types of boundary conditions at the right (outflow) boundary of the channel are considered, and we will study the influence of these on the solution. First we will prescribe a horizontal velocity $u = 1$ at the outflow boundary as well. Then we will allow a free outflow, obtained by setting $p = 0$ at the outflow boundary. The geometry and (part of) the boundary conditions are given in the figure below.



Parameter settings in the input file Use a grid without stretching with 32 cells in x-direction and 17 cells in y-direction: set `iMaxUs=34`, `jMaxUs=19` and `cx=cy=0.0`. The length of the channel is 10: set `Xmax=10`. Discretize the convection terms with a central method: set `Alpha=0.0`.
At the left (inflow) boundary the velocity is prescribed: set `left=8`, `Uin=1`, and `Vin=0`. At the upper and lower boundary no-slip conditions are imposed: set `top=2` and `bottom=2`.
Maintain the term div $\boldsymbol{u}^n$ in the Poisson equation: set `Divuv=1`. Solve the Poisson equation with an accuracy $\varepsilon = 10^{-4}$. The Poisson equation is solved with MILU. As initial solution use the solution from the previous time step: set `StrtP=1`.
Simulate the flow in the channel for 15 seconds: set `TFin=15` in the input file. Set the time step at `DelT=0.025` and the maximum allowed CFL number at `CFLMax=0.1`.
Use in the input file `PrtDt=1` and `uvp=1`, i.e. every second of the simulation output is written to screen and the velocity field is written to a file `uvpf#.dat`.

Plots for questions 5 and 6 In every question below, plot the flow at the first five seconds of the simulation and at the end of the simulation, i.e. at $t = 1, 2, \ldots, 5$ and $t = 15$. Use the post-processing menu in Matlab to plot the flow. Use the upper part of the menu to make a surface plot of the absolute velocity. Furthermore, use the middle part of the menu to make a plot of the horizontal velocity along the horizontal center line (i.e. j-plane=10). By using the '`hold`' button in the menu the plots of the horizontal velocity along the horizontal center line at $t = 1, 2, \ldots, 5$ and $t = 15$ can be put in one figure. A boundary layer and irregularities of the flow can easily be seen in the second plot.

Question 5: prescribed outflow

Prescribe a horizontal velocity $u = 1$ and $v = 0$ at the right (outflow) boundary of the channel: set `right=8`.

(a) Simulate the flow for $\nu = 0.01$. Make plots of the flow as described above. Describe and explain the development in time of the $u-$velocity along the horizontal center line.

(b) Simulate the flow for $\nu = 0.005$. Make plots of the flow as described above. Compare the flow with the one computed in (a), and explain the differences.

(c) Simulate the flow for $\nu = 0.001$. Make plots of the flow as described above.

1) Compare with the flow in (a) and (b) and explain the differences.
2) Make a surface plot of the absolute velocity in which the vector plot is included. For this, select in the menu '`arrows automatic`' instead of '`no velocity vectors`', and zoom first in on the x-interval [0,1] and then on the x-interval [9,10], i.e. take '`manual x-axis`' instead of '`automatic zoom`' and select the interval.
3) Make a plot of the $u-$velocity at the vertical section at the i-plane 32. What happens with the velocity at the upper and lower boundary?

Question 6: free outflow

Allow a free outflow condition at the right boundary of the channel: set `right=7`.

(a) (1) Simulate the flow for $\nu = 0.01$. Make plots of the flow as described above.
(2) Compare with the flow in Question 5a) and explain the differences.
(3) When the flow is fully developed (i.e. $v = 0$ and $\partial/\partial x = 0$) the $u-$velocity profile is a parabola. Compute this parabola theoretically. What is the theoretical value of the fully developed velocity at the center line $y = 0.5$? Compare this result with the numerical value. Also make a cross sectional plot of the $u-$velocity near the end of the channel to check the parabolic profile.

(b) (1) Simulate the flow for $\nu = 0.005$ and make plots of the flow.
(2) Compare with the flow in Questions 5b and 6a and explain the differences.
(3) What is the limit value of the velocity at the center line? Compare this numerical value with the theoretical value computed in the previous question.

(c) (1) Simulate the flow for $\nu = 0.001$ and make plots of the flow.
(2) Compare the flow with those computed in Questions 5c, 6a and 6b and explain the differences.
(3) What is the value of the velocity at the center line at the end of the channel? Explain why this numerical value differs from the theoretical value. Also make a cross sectional plot of the $u-$velocity near the end of the channel to see the shape of the velocity profile: is it a parabola?

(d) Repeat the simulation with a channel of respectively length 20 and length 50: set first `Xmax=20` and then `Xmax=50`. Use `TFin=50` in order to reach a steady-state solution. Also increase `Imax`, although it has little influence.
Make at the end of the simulation (i.e. at $t = 50$) a plot of the $u-$velocity along the horizontal center line (i.e. j-plane=10). What is the limit value of the $u-$velocity at the center line $y = 0.5$? Again make a cross sectional plot of the $u-$velocity near the end of the channel to see the shape of the velocity profile: is it a parabola now? Find a physical explanation of the results.

Question 5 — Lorenzo Zambelli

(a) Looking at the plots, the absolute velocity plots shows the flow in a 2D channel. At all times the highest velocity occurs in the center of the channel, while the velocity decreases towards the upper and lower walls because of the no-slip condition. Thin boundary layers are visible near the walls.

In the contour plots the max V increases from $t=1.25$ to $t=1.5$ the profile of the horizontal velocity at $y=0.5$ reveals a non physical behaviour. Instead of converging smoothly, wiggles appear near the outlet region ($x \approx 8, 10$). These oscillations become increasingly visible as $t \to \infty$.

- At $t=1$ velocity smooth
- $t=2$ & $t=3$ small oscillations
- $t=4$ & $t=5$ increase oscillations
- $t=15$ wiggles very visible and largest in amplitudes

The wiggles are caused by the convection term which it may not be sufficiently stabilized, which follows the theory on central differencing when Pe become too large → wiggles. Also, the numerical error are transported downstream and accumulate.

(b) by reducing $\nu = 0.01$ to $\nu = 0.005$ the flow become more convection-dominated.

The contour still show the aspects behaviour. However, compare to part (a):

- the wall boundary layer become thinner
- the velocity gradients near the walls become steeper
- the momentum diffuses less from the center towards the walls

which they are all consistent with lower viscosity.
The horizontal velocity at the centerline increase with time

- $u \approx 1.16$ $t=1$ → $u \approx 1.47$ $t=15$

Wiggles are still present and they propagate more upwind ($x \approx 6-10$) at $t=15$.
In this scenario the cell Pe is almost double of the cell Pe for $\nu = 0.01$, $Pe_{\nu=0.005} \approx 2\, Pe_{\nu=0.01}$ since $\nu \to 0.005$ is halve.
As discussed in the previous practice, larger Pe ⇒ convection dominated hence the diffusion is not capable enough to smooth the solution.

Lorenzo Zanbelli:

c) Following the same reasons, $\nu \to 0.001$ the flow is even more convective dominated. the effects highlighted before are stronger. the center line velocity maxima increase downstream. significant oscillations appear in the horizontal velocity profile, the wiggle even over the velocity profile no longer resembles a smooth fully developed solution.

For example, at $t=15$ when $y=0.5$ $u \approx 1.07$ at $x=0$ while $u \approx 1.35$ at $x \approx 9.5$ followed by a sharp drop.

this behavior is not physical. the velocity profile should be independent of x. this is a consequence of $Re_{\nu=0.001} \approx 10\, Re_{\nu=0.01}$

Looking at the two zoom in we can see that in $[0,1]$ the flow enters almost uniformly. the Boundary Layer start developing immediately at the walls.

Instead in $[9,10]$ the oscillations are stronger than in $[0,1]$; we have some alternating vectors directions; the velocity vary strongly from cell to cell.

In other words, this zig-zag pattern are the wiggles.

In the last plot ($T=32$) of the absolute velocity, we see a quite sharper velocity profile, with steep gradients close to the walls, with abrupt changes near the Boundaries.

the expected behavior is $u=0$ at the walls and a smooth profile between the walls, definitely not a mean direction

the drop that we see near the Boundaries indicates the lack of resolution there, hence the Boundary Layer is very thin, meaning the solution is truly sensible to discretization errors and oscillations.

Question 6

Lorenzo Zanibelli

(a) From the contour plots it is possible to see that there is a smooth transition to fully developed channel flow. Additionally:

- the flow remains symmetric respect to $y=0.5$
- thin Boundary layer (no-slip condition)
- It seem flow almost independ of x
- the velocity vector remain horizontal $\Rightarrow v \approx 0$
- Unlike Q5 no significant wiggles
- At $t=15$ the velocity seem to have converged to a steady solution
  - which can be seen from the horizontal velocity profile at $y=0.5$. In that plot it is possible to see convergence to $u \approx 1.49$

the main difference with Q5 is that the flow is smooth onto the outlet. the outflow condition $u_x=0$ & $v_x=0$ allows the flow to leave the domain without reflecting some parts back in the domain.

the maximum velocity occur in the center. the case study in this question is a classic Poiseuille flow in channel of height $H=1$, meaning the theoretical parabola is $u(y) = 4 u_{max} y(1-y)$

From computing the mean velocity by integrating $[0,1]$ we get

$$* \quad u_{max} = \frac{3}{2} U_{mean} \qquad \begin{cases} U_{mean} = \int_0^1 u(y)\,dy = C\int_0^1 y(1-y)\,dy = \frac{C}{6} \\ U_{mean}=1 \Rightarrow C=6 \end{cases}$$

$\forall 6$ at center line $u_{max} = 6(0.5)(0.5) = 1.5$

In this test problem $U_{mean}=1$ meaning $\boxed{u_{max}=1.5}$ meaning the implementation has an relative error $< 1\%$

to develop the parabola consider that $u=u(y)$ $v=0$ and $u_x=0$ · the x-momentum becomes

$$-\underbrace{p_x}_{\text{constant}} + \nu u_{yy} = 0 \quad \xrightarrow{\text{double integration}} \quad u_y = \frac{1}{\nu} p_x y + C_1, \; C_1 \in \mathbb{R}$$

$$u(y) = \frac{1}{\nu} p_x \frac{y^2}{2} + C_1 y + C_2, \; C_2 \in \mathbb{R}$$

the no slip condition imples $u(0)=u(1)=0 \Rightarrow C_2=0$ and $0 = \frac{1}{2\nu} p_x + C_1$

$$* \quad u_{max} = C\tfrac{1}{2}(1-\tfrac{1}{2}) = \frac{C}{4}$$

$$\Rightarrow \frac{u_{max}}{U_{mean}} = \frac{3}{2} \checkmark$$

$$\Rightarrow u(y) = \underbrace{\frac{1}{2\nu} p_x}_{\text{constant}} (y^2 - y) \propto C\,y(y-1) \propto C\,y(1-y)$$

Parabolic profile!

(b) Using ν to 0.005 the flow remains smooth. Comparing with $\nu = 0.01$ the lower viscosity produce thinner Boundary layer and a larger entrance region (take longer to reach stability / developed flow) the centerline approach a lower value than (a), but still close to 1.5.
the difference is mainly due to the lower viscosity.

(c) As before the flow remains smooth, no wiggles. Comparing with (a, b) the flow Developes much more slowly because smaller viscosity. As notice before lowering the viscosity increase the region needed to the flow to proper developed.

the maximum velocity is about 1.69 which is far from 1.5. $\nu = 0.001$ diffuses very slowly from the walls to the channel center. the cross section confirm this. It is not a parabola $\Leftrightarrow$ the flow is not proper developed.

(d) With 20 it is clearly not develope yet (appr $u_{max} \approx 1.4$)
At 50 instead $u_{max} \approx 1.98$. this is in line with the theory and physical interpretation. the fully Poiseuille profile forms because momentum diffuses from the walls into the channel core.
For $\nu = 0.001$ the viscous diffusion is very weak, meaning the Boundary layer grow slowly, the distance needed to develop is longer $\Leftrightarrow$ a short channel is not sufficient.

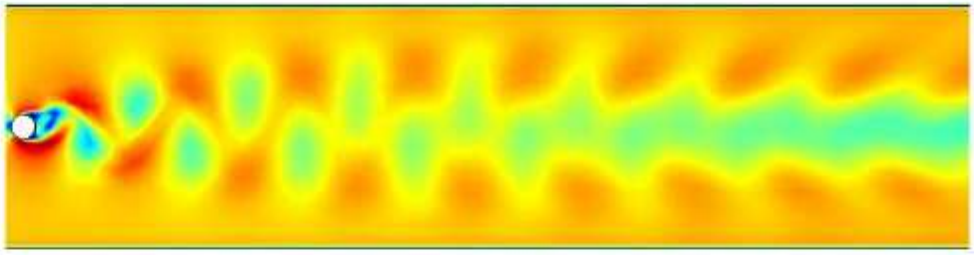
Indeed at $x = 50$ we have basically a parabola.

Bonus exercise: flow around a cylinder using COMSOL

This exercise is not an official part of the practicum. It is extra, and makes use of the commercial package COMSOL. If you do not have any prior experience with COMSOL you can simply skip the exercise.

Description In this exercise we will consider flows around a circular cylinder. The behaviour of this flow changes as the Reynolds number increases. At a certain Reynolds number boundary layer separation occurs and a wake is formed by a pair of symmetric counter-rotating vortices. This laminar wake becomes unsteady and the vortices are shed downstream forming the well-known Von Karman vortex street.

Flow configuration Consider the flow in a channel of length 10 m and height 1 m. At the left (inflow) boundary the fluid flows into the channel with a horizontal velocity u= 1 m/sec. At the upper and lower boundary of the channel we impose no-slip conditions. Place a cylinder in the channel at distance 2 m from the entry such that its center is at height 0.5 m. Its radius can be 0.05 m.
You can select water as fluid. Von Karman vortices generally start forming at low Reynolds number. In case of water, for the present geometry, we should certainly see Von Karman vortices forming at an inflow velocity of 1.25 m/sec. It is an unsteady problem, so make it run from 0 to 10 sec.
You are free to choose the spatial discretisation and time discretisation methods, taking into account wiggle and stability limits.



Questions:

(a) Try to find the Reynolds number for which Von Karman vortices start forming.

(b) We will see that a von Karman vortex affects the downstream boundaries as they do not dissipate away quickly. This may lead to inaccurate results. To mitigate this effect we can increase the size of the channel. This is not really a good option considering the fact that we are only interested in the local flow and its effects on the cylinder. Another way is to increase the viscocity of the fluid at some distance behind the cylinder. Try to do this and make the computational domain as small as possible to efficiently calculate the drag on the geometry.

(c) Try this exercise using a laminar as well as a turbulent model. Which model do you think gives a better overall picture of the phenomena and why?

Appendix: example of the input file `cfd.in`

```
***** tank geometry ************************************************
Xmin   Xmax      Ymin   Ymax
0.0    1.0       0.0    1.0

***** grid definition **********************************************
iMaxUs   jMaxUs      cx      cy         xpos   ypos
34       34          -0.95   -0.95      0.5    0.5

***** liquid properties ********************************************
Nu
0.01

***** boundary conditions and inflow characteristics ***************
left   right   top   bottom      UIn   VIn
2      2       8     2           1.0   0.0

***** discretization ***********************************************
Alpha   Divuv
0.0     1

***** poisson iteration parameters *************
Epsi     ItMax   StrtP
1.0e-3   100     1

***** time step  ********************************
TFin   DelT    CFLMax
25     0.025   0.5

***** print/plot control *******************************************
PrtDt  uvp
1      0


-----------------------------------------------------------------------
***** E X P L A N A T I O N ************* E X P L A N A T I O N *****
-----------------------------------------------------------------------

**TANK GEOMETRY**
Xmin, Xmax, Ymin, Ymax : position of boundaries of computational domain

**GRID DEFINITION**
iMaxUs (<=130), jMaxUs (<=130) : number of cells including mirror cells,
                                 number of 'real' cells is iMaxUs-2, jMaxus-2
cx, cy     : stretchparameters - 0=no stretch;
                                 >0=smaller cells near position
                                    indicated by xpos,ypos;
                                 <0=smaller cells away from xpos,ypos
```

```
**LIQUID PROPERTIES**
Nu         : kinematic viscosity

**BOUNDARY CONDITIONS AND INFLOW CHARACTERISTICS**
left,right,top,bottom  : type of boundary condition
                              1=slip
                              2=no-slip
                              7=free outflow : p=0 as boundary condition
                              8=inflow  : the velocities are prescribed
UIn,VIn    : in case of inflow conditions, prescribed velocities
             the sign of UIn/VIn corresponds to the x/y-direction (hence it
             does not necessarily correspond to the inward pointing direction)

**DISCRETIZATION**
Alpha      : upwind parameter, discretization of convection term
                0 = central
                1 = upwind
Divuv      : treatment of divergence
                0 = substitute div u^n=0 in Poisson equation
                1 = maintain div u^n in Poisson equation

**POISSON ITERATION PARAMETERS**
Epsi       : convergence criterion of Poisson solver
ItMax      : maximal allowed number of iterations of Poisson solver
Strtp      : initial field of Poisson solver
               0 = zero field p=0
               1 = solution from previous time step, p=poud;

**TIME STEP**
TFin       : endtime of simulation
DelT       : timestep (is adjusted during the simulation in order to
             keep the CFL number smaller than CFLMax)
CFLMax     : maximum allowed CFL number

** PRINT/PLOT CONTROL**
PrtDt      : time between 2 small printouts (to the screen and a file)
uvp        : velocity and pressure in Matlab format
               0=only at end of computation (preferable)
               1=at every small printout (takes a lot of memory)
```